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A market-rule-informed neural network for efficient imbalance electricity price forecasting

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ABSTRACT

Accurate and efficient imbalance electricity price forecasting is critical for industrial energy trading in balancing markets. Battery assets and automated trading systems require forecasts with limited computational resources and near-real-time speed. The imbalance price is formed through known market rules that aggregate various market signals into the final market outcome. For accurate price forecasting, this raises a methodological question: can future prices be sufficiently forecast from only lagged prices, or are raw market signals needed? This paper argues for the latter and proposes a market-rule-informed neural network (MRINN) that embeds market rules into the latent space. By embedding these rules, the proposed model preserves raw signal information while avoiding the need to relearn known market rules, leading to a compact model with faster training and inference. We further examine operational robustness under delayed or missing signals and characterize performance scaling across input length and forecasting horizon. Our case studies provide empirical evidence that the proposed MRINN improves forecasting accuracy by 8% compared with the best lagged-price-only baseline, while reducing the number of trainable parameters by around 90% relative to raw-signal neural baselines. The data and code are open-sourced at https://github.com/runyao-yu/MRINN.

1. Introduction

Electricity markets play a central role in modern power systems by coordinating generation scheduling, flexibility procurement, and real-time system balancing. In most liberalized systems, trading is organized sequentially. The day-ahead market clears the bulk of energy transactions one day before delivery, the intraday market allows participants to update positions closer to real time, and the balancing market resolves residual imbalances during system operation. In the last sequence, the balancing market participants may provide balancing services through accepted bids and offers, or hold imbalanced positions that are settled at the imbalance price [1].

Imbalance prices are determined by predefined market rules that map observable market signals, such as activated reserve energy prices and system imbalance volumes, to an imbalance price. However, these rules are not universal: different countries and bidding zones, such as Germany and Austria, adopt different market rules. Although, the pricing formation is explicitly specified by these rules, forecasting imbalance prices remains challenging due to nonlinear and piecewise rule structures, scarcity adjustments, and rapidly changing operating conditions [2–4].

Recent advances in machine learning have stimulated a growing body of work on nonlinear electricity price forecasting. Most studies focus on day-ahead price forecasting, where advanced machine learning and interpretable methods have been developed and evaluated [11–16]. Beyond the day-ahead market, recent intraday price forecasting studies have increasingly exploited high-frequency market information, including micro-orderbook-based representations [17,18]. However, compared with day-ahead and intraday price forecasting, imbalance price forecasting remains relatively underexplored. As summarized in Table 1, most existing studies rely on conventional deep learning models, such as Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM) [3–5,7,9,10]. A more advanced variant, Attention-based Bidirectional LSTM (AttnBiLSTM), further exploits sequential dependencies and attention mechanisms to improve forecasting accuracy [6,8]. However, most existing approaches consider imbalance

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Table 1
Summary of imbalance price forecasting studies, including the deep learning model used, whether probabilistic forecasts are produced, whether market rules are incorporated, the studied country, and the year.

Study	Model	Probabilistic	Market rules	Country	Year
Dumas et al. [5]	MLP	✓	✗	Belgium	2019
Bottieau et al. [6]	AttnBiLSTM	✓	✗	Belgium	2022
Narajewski [3]	MLP	✓	✗	Germany	2022
Ganesh and Bunn [7]	MLP/LSTM/ Transformer	✓	✗	UK	2023
O'Connor et al. [4]	MLP/LSTM	✗	✗	Ireland	2024
Deng et al. [8]	AttnBiLSTM	✗	✗	UK	2024
Plakas et al. [9]	LSTM	✓	✗	Greece	2025
O'Connor et al. [10]	MLP/LSTM	✓	✗	Ireland	2025

prices as generic time-series targets and rely on purely data-driven models. In contrast to many other domains, imbalance price formation is governed by transparent and explicitly stated market rules. Surprisingly, much of the current literature does not exploit this structural information, implicitly requiring models to **relearn** deterministic mappings that are already known. This gap motivates a re-examination of how market-rule knowledge can inform forecasting models.

Two forecasting approaches can be distinguished: one uses only lagged imbalance prices as input, while the other uses raw market signals from which imbalance prices are formed. Forecasting only from lagged prices implicitly assumes that the historical imbalance price provides a sufficient representation of the underlying market state. However, this assumption is restrictive as the lagged imbalance price is only a compressed outcome of that state. This corresponds to a **non-invertible** mapping problem, where different combinations of raw market signals may lead to the same observed imbalance price. Therefore, a model that only observes the lagged imbalance price may not distinguish operationally different situations. In contrast, the raw features retain information on balancing activation, market-reference prices, and system imbalance. Not processing such features and failing to distinguish these operational scenarios can lead to degraded forecasting performance, as we later show empirically.

Moreover, when using raw features, a key question is how to effectively exploit prior knowledge encoded in the market rules. To this end, we design a Market-Rule-Informed Neural Network (MRINN) that takes raw features as input and embeds market rules as differentiable operators in the latent space. This design injects structural inductive bias while avoiding information loss due to premature compression. This rule-based injection also prevents the model from **relearning** price formation rules solely from data through heavy parameterization. As a lightweight model, MRINN is practically relevant for algorithmic trading in balancing markets. In such settings, forecasting models may need to be embedded into local decision-making pipelines for real-time trading [1,19], site-level server deployment and local optimization of distributed energy resources [20], and edge-based battery bidding [21]. In these applications, memory and computational resources are critical constraints.

Furthermore, imbalance price forecasting often relies on heterogeneous real-time signals, including system imbalance, balancing activation features, and short-term market indices [7,9,22]. In balancing markets, the timely availability of system-imbalance information is economically important as delays in information transparency can affect real-time statistical-arbitrage positions [1]. Existing studies commonly evaluate forecasting models under the assumption that these input signals are fully observed and temporally aligned at the forecasting time. In practical deployments, however, this assumption may not always hold as market signals may be affected by publication lags, communication delays, or measurement outages [23]. As a result, not all input signals are reliably available in real time. Thus, it remains unclear how forecasting performance degrades when specific price-component information is unavailable at the forecasting time.

In addition, prior studies have investigated how forecasting performance changes with input length and forecasting horizon, often referred to as empirical scaling laws in time-series forecasting [2,8]. However, these analyses are typically conducted for generic data-driven forecasting models without considering market rules. When market rules inform the forecasting model, the effect of input length and forecasting horizon may no longer follow the same pattern, as the model relies not only on temporal information but also on rule-derived price components. Therefore, it remains unclear whether similar scaling law persists in market-rule-informed forecasting. In this paper, we systematically analyze operational robustness under different data-availability settings and characterize the scaling law of MRINN with respect to input length and forecasting horizon.

Our main contributions are:

- **We formulate imbalance price forecasting as a non-invertible problem.** Empirical results show that lagged-price compression causes information loss, motivating the use of raw market signals rather than lagged prices alone.
- **We propose MRINN, a parameter-efficient neural network.** The proposed MRINN takes raw market signals as input and is informed by market rules through differentiable latent-space operators. This reduces the need for the model to relearn known rules from data.
- **We provide practical engineering insights for industrial deployment.** Through price-component ablation and empirical scaling-law analysis, we quantify how component availability, input length, and forecasting horizon affect predictive performance.

The remainder of this paper is organized as follows. Section 2 introduces the dataset, providing generic insights. Section 3 explains the imbalance market rules, which are subsequently injected into the proposed MRINN model in Section 4. Section 5 presents case studies. Section 6 concludes the paper.

2. Data

The dataset has a 15-min resolution and spans 4 years from 2022-01-01 to 2026-01-01. Fig. 1 summarizes the empirical characteristics of the key features from the Austrian balancing market.

2.1. Feature categories

The input features are categorized into three groups according to the market rules, which will be introduced in Section 3. These three categories correspond to three intermediate price components used to form the imbalance price P_t.

The first group contains balancing-energy features, including automatic Frequency Restoration Reserve (aFRR) and manual Frequency Restoration Reserve (mFRR) activation volumes and corresponding activation prices. Although aFRR and mFRR are European balancing-service products, analogous real-time balancing mechanisms exist in other systems under different terminology and market arrangements. These variables are directly related to the balancing-energy price component P_t^{bal}.

The second group contains market-reference features, including day-ahead prices, intraday 15-min and 60-min price indices, and the corresponding liquidity measures. These variables construct the market-reference component P_t^{mkt}.

The third group contains system imbalance V_t, which falls under the scarcity-function feature category. This variable determines both the direction and magnitude of the scarcity component P_t^{scar}.

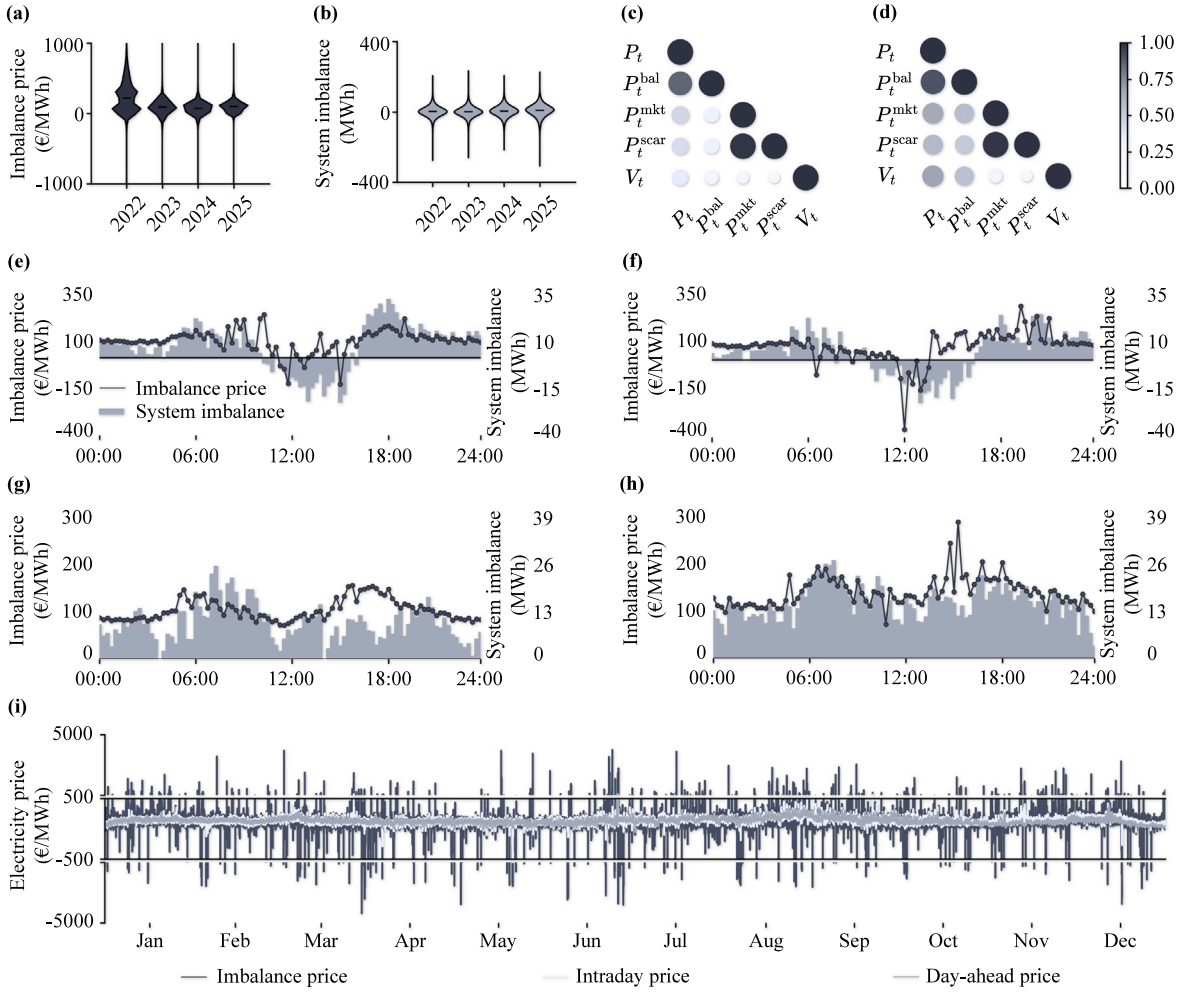


Fig. 1. Imbalance data analysis. **(a)** Yearly distributions of imbalance prices from 2022 to 2025. **(b)** Yearly distributions of system imbalance from 2022 to 2025. **(c)** Pearson correlation magnitude among the imbalance price, the three price components, and system imbalance. **(d)** Spearman correlation magnitude among the same variables. **(e)–(h)** Average imbalance price and system imbalance across delivery times for spring, summer, autumn, and winter, respectively. **(i)** Comparison of day-ahead, intraday, and imbalance prices averaged across 2022–2025.

2.2. Exploratory data analysis

We conduct an exploratory data analysis to characterize the distributional, correlation, seasonal, and volatility patterns of the imbalance-price data.

Temporal evolution of imbalance features: Fig. 1**(a)** shows that the distribution of imbalance prices has pronounced long tails, especially during the energy crisis period in 2022, before becoming more stable in later years. In contrast, Fig. 1**(b)** shows that system imbalance has smaller deviations. Its distribution also changes across years, with a slightly more negative imbalance pattern in 2025, reflecting increased renewable integration.

Correlation between imbalance features: Fig. 1**(c)** and **(d)** report Pearson and Spearman correlation magnitudes among P_t, the three price components, and V_t. The Pearson correlations in Fig. 1**(c)** show that P_t is most strongly linearly associated with P_t^{bal}, while its linear dependence on P_t^{mkt}, P_t^{scar}, and V_t is weaker. In contrast, the Spearman correlations in Fig. 1**(d)** are stronger across most variable pairs, indicating nonlinear dependence induced by the imbalance market rules.

Seasonal delivery-time patterns: Fig. 1**(e)–(h)** show average imbalance prices and system imbalance across delivery times for spring, summer, autumn, and winter, respectively, in 2025. The seasonal profiles reveal clear patterns: spring and summer show similar ranges, with summer exhibiting strong negative price movements around noon, while autumn and winter are more associated with positive imbalance-price levels, especially during afternoon and evening periods. These seasonal patterns motivate an evaluation protocol whose testing periods collectively cover a full year, ensuring that model performance is assessed across different seasonal conditions.

Volatility of imbalance prices: Fig. 1**(i)** compares day-ahead, intraday, and imbalance prices averaged across 2022–2025. The observed volatility increases from day-ahead to intraday prices and is highest for imbalance prices. In particular, imbalance prices exhibit frequent extreme spikes in both directions, reflecting stronger volatility and tail risk. This highlights the difficulty of imbalance price forecasting and motivates the use of probabilistic models that can capture extreme price movements.

3. Imbalance market rules

This section introduces the imbalance market rules that map the raw feature vector F_t into the imbalance price P_t. These rules provide the structural prior that will be embedded into the proposed MRINN model in Section 4.

Let $F_t^{(k)} \in \mathbb{R}$ denote the value of the kth input feature at time step t, where $k \in \{1, \dots, K\}$ indexes the feature type and K is the total number of input features. For example, $F_t^{(1)}$ may represent the system imbalance

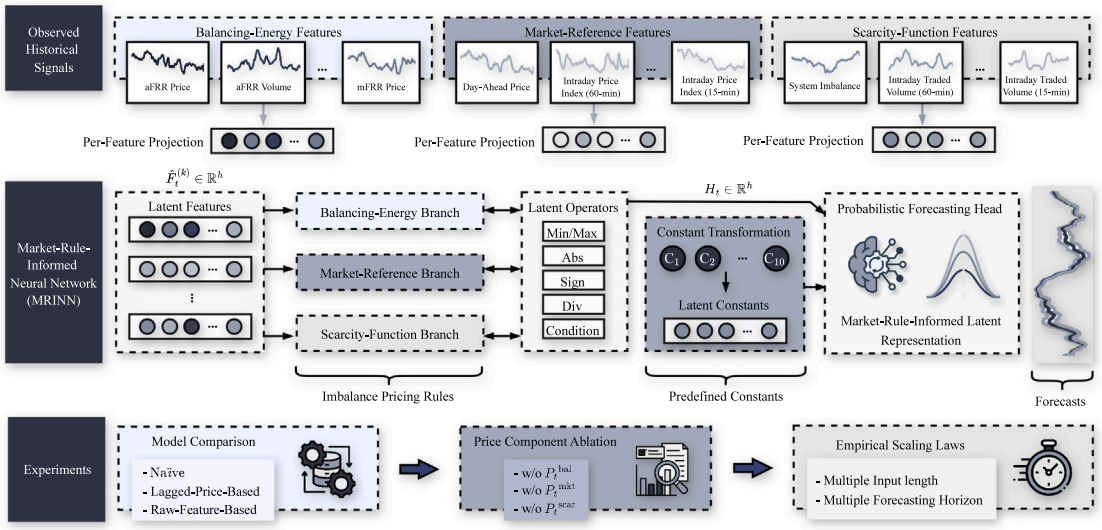


Fig. 2. Structure illustration of the proposed MRINN and the corresponding experimental analysis.

V_t. We consider quarter-hourly settlement periods. The imbalance price P_t is defined as the extremum over three price components, with the choice of $\min(\cdot)$ or $\max(\cdot)$ determined by the sign of V_t:

$$P_t = \begin{cases} \min\left(P_t^{\text{bal}},\ P_t^{\text{mkt}},\ P_t^{\text{scar}}\right), & V_t < 0, \\ \max\left(P_t^{\text{bal}},\ P_t^{\text{mkt}},\ P_t^{\text{scar}}\right), & V_t \geq 0, \end{cases} \tag{1}$$

where P_t^{bal} is the balancing-energy-based price component (Eq. (2)), P_t^{mkt} is an exchange-based market reference price component (Eq. (7)), and P_t^{scar} is a scarcity-function price component (Eq. (15)). Each component is specified in the following subsections (A–C).

3.1. Balancing-energy price component

This component links the imbalance price to the balancing-energy outcome in period t and depends on the activation direction and the sign of the system imbalance V_t:

$$P_t^{\text{bal}} = \begin{cases} P_t^{\text{act},+}, & I_t^+ = 1,\ I_t^- = 1,\ V_t \geq 0, \\ P_t^{\text{act},-}, & I_t^+ = 1,\ I_t^- = 1,\ V_t < 0, \\ P_t^{\text{act},+}, & I_t^+ = 1,\ I_t^- = 0, \\ P_t^{\text{act},-}, & I_t^+ = 0,\ I_t^- = 1, \\ P_t^{\text{VoAA},+}, & I_t^+ = 0,\ I_t^- = 0,\ V_t \geq 0, \\ P_t^{\text{VoAA},-}, & I_t^+ = 0,\ I_t^- = 0,\ V_t < 0. \end{cases} \tag{2}$$

where $P_t^{\text{act},+}$ and $P_t^{\text{act},-}$ denote the activation-weighted balancing-energy prices, as defined in Eqs. (3)–(4). $P_t^{\text{VoAA},+}$ and $P_t^{\text{VoAA},-}$ denote the values of avoided activation (VoAA) and are computed as the lowest and highest offer prices in the local positive and local negative aFRR balancing-energy merit-order lists for quarter-hour t, respectively. I_t^+ and I_t^- indicate whether positive or negative activation occurs, as defined in Eqs. (5)–(6).

$$P_t^{\text{act},+} = \frac{E_t^{\text{aFRR},+} P_t^{\text{aFRR},+} + E_t^{\text{mFRR},+} P_t^{\text{mFRR},+}}{E_t^{\text{aFRR},+} + E_t^{\text{mFRR},+}}. \tag{3}$$

$$P_t^{\text{act},-} = \frac{E_t^{\text{aFRR},-} P_t^{\text{aFRR},-} + E_t^{\text{mFRR},-} P_t^{\text{mFRR},-}}{E_t^{\text{aFRR},-} + E_t^{\text{mFRR},-}}. \tag{4}$$

$$I_t^+ = \mathbb{I}\left[E_t^{\text{aFRR},+} + E_t^{\text{mFRR},+} > 0\right]. \tag{5}$$

$$I_t^- = \mathbb{I}\left[E_t^{\text{aFRR},-} + E_t^{\text{mFRR},-} > 0\right]. \tag{6}$$

where $E_t^{\text{aFRR},\pm}$ and $E_t^{\text{mFRR},\pm}$ are the activated aFRR and mFRR balancing-energy volumes (MWh), respectively. $P_t^{\text{aFRR},\pm}$ and $P_t^{\text{mFRR},\pm}$ are the corresponding prices (€/MWh). $\mathbb{I}[\cdot]$ is the indicator function.

3.2. Market-reference price component

This component constructs an exchange-based reference for quarter-hour t by forming a liquidity-dependent weighted sum of ramp-adjusted intraday and day-ahead indices:

$$P_t^{\text{mkt}} = w_t^{\text{ID15}}\, P_t^{\text{ID15,mkt}} + w_t^{\text{ID60}}\, P_t^{\text{ID60,mkt}} + w_t^{\text{DA}}\, P_t^{\text{DA,mkt}}. \tag{7}$$

where $P_t^{\text{ID15,mkt}}$, $P_t^{\text{ID60,mkt}}$, and $P_t^{\text{DA,mkt}}$ are defined in (8)–(10), and the weighting factors w_t^{ID15}, w_t^{ID60}, and w_t^{DA} are defined in (12)–(14).

$$P_t^{\text{ID15,mkt}} = P_t^{\text{ID15}} + \text{ramp}(V_t) \cdot \max\left(\text{C}_1,\ \text{C}_0\,|P_t^{\text{ID15}}|\right). \tag{8}$$

$$P_t^{\text{ID60,mkt}} = P_t^{\text{ID60}} + \text{ramp}(V_t) \cdot \max\left(\text{C}_2,\ \text{C}_0\,|P_t^{\text{ID60}}|\right). \tag{9}$$

$$P_t^{\text{DA,mkt}} = P_t^{\text{DA}} + \text{ramp}(V_t) \cdot \max\left(\text{C}_3,\ \text{C}_0\,|P_t^{\text{DA}}|\right). \tag{10}$$

where P_t^{ID15} is the intraday price index for 15-min products, P_t^{ID60} is the corresponding intraday price index for 60-min products, and P_t^{DA} is the day-ahead market coupling price (all in €/MWh). The constants $\text{C}_0 = 0.1$, $\text{C}_1 = 5$ (€/MWh), $\text{C}_2 = 10$ (€/MWh), and $\text{C}_3 = 15$ (€/MWh) are predefined parameters, and the ramp function is defined as:

$$\text{ramp}(V_t) = \begin{cases} -1, & V_t < -\text{C}_4, \\ \dfrac{V_t}{\text{C}_4}, & -\text{C}_4 \leq V_t \leq \text{C}_4, \\ +1, & V_t > \text{C}_4. \end{cases} \tag{11}$$

where $\text{C}_4 = 50$ (MW) is another constant and controls the width of the linear transition around $V_t \approx 0$.

$$w_t^{\text{ID15}} = \min\left(1,\ \frac{L_t^{\text{ID15}}}{\text{C}_5}\right), \tag{12}$$

$$w_t^{\text{ID60}} = \min\left(1 - w_t^{\text{ID15}},\ \frac{L_t^{\text{ID60}}}{\text{C}_6}\right), \tag{13}$$

$$w_t^{\text{DA}} = 1 - w_t^{\text{ID15}} - w_t^{\text{ID60}}, \tag{14}$$

where L_t^{ID15} and L_t^{ID60} denote the traded volumes used to assess intraday liquidity, and $\text{C}_5 = 200$ (MW) and $\text{C}_6 = 200$ (MW) are predefined constants for liquidity indication.

3.3. Scarcity-function price component

This component introduces a scarcity adjustment as a signed cubic function of the absolute system imbalance $|V_t|$:

$$P_t^{\text{scar}} = \begin{cases} P_t^{\text{base}}, & |V_t| \le \mathrm{C}_7, \\ P_t^{\text{base}} + \operatorname{sgn}(V_t)\,\mathrm{C}_{10}\left(\dfrac{|V_t| - \mathrm{C}_7}{\mathrm{C}_9 - \mathrm{C}_7}\right)^3, & \mathrm{C}_7 < |V_t| \le \mathrm{C}_8, \\ P_t^{\text{base}} + \operatorname{sgn}(V_t)\,\mathrm{C}_{10}\left(\dfrac{\mathrm{C}_8 - \mathrm{C}_7}{\mathrm{C}_9 - \mathrm{C}_7}\right)^3, & |V_t| > \mathrm{C}_8. \end{cases} \tag{15}$$

where $\operatorname{sgn}(V_t) \in \{-1, +1\}$ denotes the sign of the system imbalance. $\mathrm{C}_7 = 200$ (MW) is the deadband threshold (no scarcity adjustment for $|V_t| \le \mathrm{C}_7$), $\mathrm{C}_8 = 800$ (MW) is the cap threshold beyond which the scarcity adjustment saturates, $\mathrm{C}_9 = 1000$ (MW) is the imbalance intersection point shaping the cubic curve, and $\mathrm{C}_{10} = 1000$ (€/MWh) is the price intersection point scaling the magnitude of the scarcity adjustment. P_t^{base} is an *unramped* exchange reference price, defined similarly to (7) as:

$$P_t^{\text{base}} = w_t^{\text{ID15}}\, P_t^{\text{ID15}} + w_t^{\text{ID60}}\, P_t^{\text{ID60}} + w_t^{\text{DA}}\, P_t^{\text{DA}}. \tag{16}$$

4. Market-rule-informed neural network

Since the imbalance price is generated by the known market rules described in Section 3, directly compressing the raw features into the lagged imbalance price may discard predictive information. We denote the full raw feature vector as $F_t = [F_t^{(1)}, F_t^{(2)}, \ldots, F_t^{(K)}]$, and write the market-rule mapping compactly as $P_t = g(F_t)$.

This mapping is generally **non-invertible** and acts like a lossy compression: different raw feature states can lead to the same imbalance price. For example,

$$\begin{aligned} F_t = [1, 0, 1, \ldots] &\xrightarrow{g(\cdot)} P_t = 100 \xrightarrow{\text{model}} P_{t+15\text{min}} = 200, \\ F_t = [1, 2, 3, \ldots] &\xrightarrow{g(\cdot)} P_t = 100 \xrightarrow{\text{model}} P_{t+15\text{min}} = 500. \end{aligned} \tag{17}$$

If a forecasting model only uses P_t as input, these two cases become indistinguishable: the model receives the same input value, $P_t = 100$, but is expected to predict two different future prices. As a result, such a model is likely to learn an averaged prediction. To preserve the original signal content while still exploiting market-rule priors, we propose MRINN. The model takes raw features as inputs and embeds the known market rules as differentiable operators in the latent space.

4.1. Backbone

We first project each raw feature into a latent space of dimension h:

$$F_t^{(k)} \in \mathbb{R} \xrightarrow{\text{Project}} \hat{F}_t^{(k)} \in \mathbb{R}^h. \tag{18}$$

After projection, each feature is represented as a latent vector, namely $\hat{F}_t^{(1)}, \hat{F}_t^{(2)}, \ldots, \hat{F}_t^{(K)}$. Accordingly, the market rules are extended from **scalar operations** to **vector operations**. In the latent space, these feature representations are passed through differentiable pricing-rule operators to obtain a latent representation of the imbalance price $H_t \in \mathbb{R}^h$, as illustrated in Fig. 2. However, when the market rules are implemented in the latent space, several of its original operations become challenging for backpropagation, as detailed in the following Sections 4.1.1–4.1.6.

4.1.1. Min/Max block

The min and max operators are non-smooth, which may obstruct gradient flow during backpropagation. Motivated by prior work that uses softplus to impose smooth functional constraints in neural networks [24], we approximate the min and max with the softplus function.

For two latent vectors $a, b \in \mathbb{R}^h$, note that the max operator can be rewritten as

$$\max(a, b) = b + \max(a - b, 0). \tag{19}$$

We therefore approximate the hard $\max(\cdot, 0)$ operation with the softplus function, where

$$\operatorname{softplus}(z) = \log(1 + e^z). \tag{20}$$

This gives the latent-space max block:

$$\max_{\text{latent}}(a, b) = b + \operatorname{softplus}(a - b). \tag{21}$$

Similarly, the latent-space min block is defined by symmetry as

$$\min_{\text{latent}}(a, b) = -\max_{\text{latent}}(-a, -b). \tag{22}$$

All operations are applied element-wise. This yields differentiable and smooth approximations of the extremum operators appearing in Eqs. (1), (8), (9), (10), (12), and (13).

4.1.2. Absolute function block

As the absolute function $|\cdot|$ is non-differentiable at zero, we use the following differentiable approximation:

$$\operatorname{abs}_{\text{latent}}(a) = \sqrt{a^2 + \epsilon}, \tag{23}$$

where $\epsilon = 10^{-7}$. This yields a differentiable approximation of abs used in Eqs. (8), (9), (10), and (15).

4.1.3. Sign function block

As the $\operatorname{sign}(\cdot)$ operator is discontinuous at zero, we approximate it using the hyperbolic tangent function:

$$\operatorname{sign}_{\text{latent}}(a) = \tanh(a), \tag{24}$$

This function is fully differentiable and smoothly saturates toward +1 and −1, thereby mimicking the behavior of the original sign operator in Eq. (15).

4.1.4. Division block

As division can become numerically unstable when the denominator approaches zero, we add a small positive constant to the denominator:

$$\operatorname{div}_{\text{latent}}(a, b) = \frac{a}{b + \epsilon}, \tag{25}$$

This yields a numerically stable approximation of the division operation in Eqs. (3) and (4) within the latent space.

4.1.5. If–else condition block

Unlike decision trees or rule-based systems, standard neural networks do not naturally support hard if–else conditions. Inspired by deep decision forests [25], we replace discrete branching with a differentiable soft-selection mechanism based on softmax weights. Let us consider the following exemplary hard conditional rule:

$$\operatorname{cond}(A, B, C; a, b, c) = \begin{cases} A, & \text{if } a > b, \\ B, & \text{if } b > c, \\ C, & \text{otherwise}, \end{cases} \tag{26}$$

where $a, b, c \in \mathbb{R}^h$ denote condition-related variables and $A, B, C \in \mathbb{R}^h$ denote the corresponding candidate outputs. To obtain a differentiable approximation, we first concatenate the condition variables and map them to branch-selection weights:

$$(w_1, w_2, w_3) = \operatorname{softmax}\big(\phi([a; b; c])\big), \tag{27}$$

where $\phi(\cdot)$ denotes a learnable transformation, $w_1, w_2, w_3 \in [0, 1]$, and

$$w_1 + w_2 + w_3 = 1. \tag{28}$$

The hard if–else rule is then replaced by the following latent-space condition block:

$$\mathrm{cond}_{\mathrm{latent}}(A, B, C; a, b, c) = A \odot w_1 + B \odot w_2 + C \odot w_3, \tag{29}$$

where $\odot$ denotes element-wise multiplication. Given different inputs a, b, c, the softmax function learns to adapt the branch probabilities accordingly. For example, if the first branch is strongly favored, the weights may approach $w_1 \approx 1$ and $w_2, w_3 \approx 0$, so that $\mathrm{cond}_{\mathrm{latent}}(A, B, C; a, b, c)$ effectively passes only A to the output. In this way, the original hard conditional logic is approximated by a differentiable block in the latent space, as used in Eqs. (1), (2), (11), and (15).

4.1.6. Constant transformation block

As the market rule is implemented in the latent space, the predefined constants $C_1, \ldots, C_{10}$ from Section 3 cannot be directly used. First, they are scalar quantities, whereas the latent-space pricing mechanism operates on vectors in $\mathbb{R}^h$. Second, they are specified in the original physical scale, whereas the corresponding input features are scaled before entering the latent representation. To ensure consistency, we group training features with the same physical unit, concatenate them along the sample dimension, and fit a `RobustScaler` for each unit group using the training data only. Each predefined constant is then transformed using the scaler associated with its corresponding unit. Finally, the transformed constant is broadcast to dimension h so that it can be used in latent-space vector operations.

4.2. Head

To obtain reliable probabilistic forecasts, we equip MRINN with the hierarchical quantile head proposed in [26], which is designed to mitigate quantile crossing. The final latent representation $H_t \in \mathbb{R}^h$ is fed into a dense layer to produce the median forecast:

$$\hat{y}_{t,0.50} = f_{0.50}(H_t). \tag{30}$$

The remaining quantiles are then constructed hierarchically through non-negative increments. For example, for the quantile set $Q = \{0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90\}$, the lower quantiles are generated outward from the median as

$$\begin{aligned} \hat{y}_{t,0.45} &= \hat{y}_{t,0.50} - \mathrm{softplus}(f_{0.45}(H_t)), \\ \hat{y}_{t,0.25} &= \hat{y}_{t,0.45} - \mathrm{softplus}(f_{0.25}(H_t)), \\ \hat{y}_{t,0.10} &= \hat{y}_{t,0.25} - \mathrm{softplus}(f_{0.10}(H_t)). \end{aligned} \tag{31}$$

Similarly, the upper quantiles are generated outward from the median as

$$\begin{aligned} \hat{y}_{t,0.55} &= \hat{y}_{t,0.50} + \mathrm{softplus}(f_{0.55}(H_t)), \\ \hat{y}_{t,0.75} &= \hat{y}_{t,0.55} + \mathrm{softplus}(f_{0.75}(H_t)), \\ \hat{y}_{t,0.90} &= \hat{y}_{t,0.75} + \mathrm{softplus}(f_{0.90}(H_t)). \end{aligned} \tag{32}$$

where $f_\tau(\cdot)$ denotes a quantile-specific dense layer. Since each increment is constrained to be non-negative, the resulting forecasts satisfy the desired ordering $\hat{y}_{t,0.10} \le \hat{y}_{t,0.25} \le \hat{y}_{t,0.45} \le \hat{y}_{t,0.50} \le \hat{y}_{t,0.55} \le \hat{y}_{t,0.75} \le \hat{y}_{t,0.90}$, thereby preventing quantile crossing and improving the reliability of probabilistic forecasts.

4.3. Loss

Following prior works [26], we optimize the model using the Average Quantile Loss (AQL). Since imbalance prices are highly volatile, quantile loss is used to evaluate probabilistic forecasts, where quantiles such as 0.1 and 0.9 describe the lower and upper ranges of possible future prices. Specifically, AQL measures the accuracy of multiple predicted quantiles:

$$\mathrm{AQL} = \frac{1}{N|Q|} \sum_{i=1}^{N} \sum_{\tau \in Q} L_\tau(y_i, \hat{y}_{i,\tau}), \tag{33}$$

Table 2
Three-fold rolling evaluation protocol.

Splitting	Fold 1	Fold 2	Fold 3
Training	2022-01–2024-08	2022-01–2024-12	2022-01–2025-04
Validation	2024-09–2024-12	2025-01–2025-04	2025-05–2025-08
Testing	2025-01–2025-04	2025-05–2025-08	2025-09–2025-12

where y_i and $\hat{y}_{i,\tau}$ denote the true price and the predicted price quantile at level τ, respectively. The quantile loss L_τ is defined as:

$$L_\tau(y_i, \hat{y}_{i,\tau}) = \begin{cases} \tau \cdot (y_i - \hat{y}_{i,\tau}), & \text{if } y_i \ge \hat{y}_{i,\tau}, \\ (1-\tau) \cdot (\hat{y}_{i,\tau} - y_i), & \text{otherwise.} \end{cases} \tag{34}$$

5. Case study

5.1. Case study settings

5.1.1. Model comparison

We benchmark against three categories of baselines: (i) naïve baselines, (ii) lagged-price-based baselines, and (iii) raw-feature-based baselines. The hyperparameter optimization is described in Appendix A.2. The forecasting process follows a 15-minute rolling window. To ensure strict temporal causality and avoid data leakage, only information available up to forecasting time t is used. We adopt a three-fold rolling evaluation protocol (Table 2). The testing periods of the three folds collectively cover a full year. To ensure robustness, each reported result is averaged over five random seeds and three folds for each model.

Naïve Models. We include three Naïve Baselines. The first adopts the latest imbalance price P_t as the forecast for the next period $P_{t+15\mathrm{min}}$, corresponding to a persistence baseline. In addition, prior studies have identified the 15-min intraday price index[1] as a strong persistence benchmark [3]. For consistency, we also include 60-min index as an additional naïve benchmark.

Lagged-Price-Based Models. When using only the lagged imbalance price P_t as input, neither explicit temporal nor spatial inductive bias from advanced models is required. We therefore consider Linear Quantile Regression (LQR) for linear modeling, and XGBoost (XGB) and Multi-Layer Perceptron (MLP) for non-linear modeling, where prior work reports that XGB outperforms both persistence benchmarks and LSTM-based models [9], while another study shows that MLP achieves lower testing loss than Gated Recurrent Unit (GRU), Temporal Fusion Transformer (TFT), and N-BEATS [7].

Raw-Feature-Based Models. When using raw input signals, more advanced deep learning models can be considered. In the context of imbalance price forecasting, prior studies report that the Attention-Based Bidirectional LSTM (AttnBiLSTM) outperforms several competitive baselines, including Auto-Regressive Moving Average (ARMA), Gradient Boosting Regression Tree (GBRT), and Bahdanau-Based Sequence-to-Sequence (B-Seq2Seq) [6]. Similarly, another study also shows that this variant also outperforms Temporal Convolutional Network (TCN), LightGBM, and Transformer-based models [8]. In addition, we include several recent advanced deep learning models that have achieved strong performance across a wide range of domains, including transportation, weather, and electricity, namely iTransformer [27], PatchTST [28], TimesNet [29], and TimeXer [30].

[1] In our setting, the index is defined as the volume-weighted average price over the 3-hour interval before delivery. Specifically, the 15-min and 60-min indices are computed from the 15-min and 60-min products, respectively.

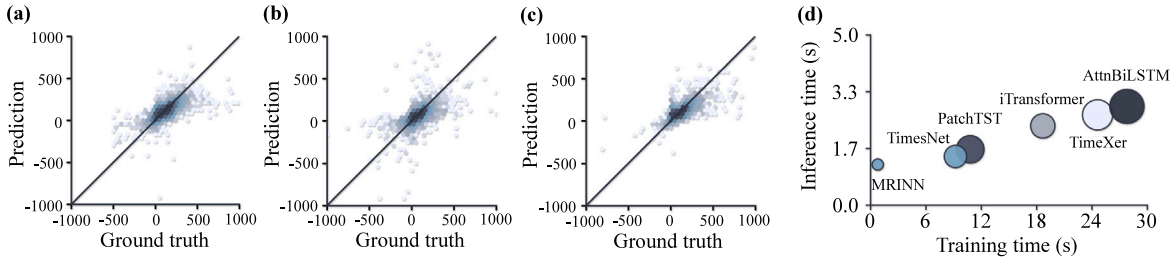


Fig. 3. Forecasting and computational performance comparison. **(a)–(c)** show predicted versus observed imbalance prices across the three testing folds. **(d)** compares inference time and training time per epoch using CPU. The bubble size represents the number of trainable parameters.

5.1.2. Price component ablation

As described in Section 3, the imbalance price in Eq. (1) is determined by three price components: P_t^{bal}, P_t^{mkt}, and P_t^{scar}. Although these components jointly determine the final imbalance price, their relative contribution to forecasting is not necessarily equal. Therefore, it is important to understand how each rule-derived price component, together with its associated input features, contributes to the predictive representation learned by the model. To this end, we conduct a component-removal analysis. For each experiment, we remove one price-component branch from the latent market rules and use the remaining components to forecast the future price.

(i) Without balancing-energy component. The aFRR- and mFRR-related input features are removed, and the market rules in Eqs. (2)–(6) are not used to construct the latent representation of P_t^{bal}. The model therefore relies only on the latent representations of P_t^{mkt} and P_t^{scar} to forecast $P_{t+15\text{min}}$.

(ii) Without market-reference component. The market-related price features are removed, and the market rules in Eqs. (7)–(14) are not used to construct the latent representation of P_t^{mkt}. The model therefore relies only on the latent representations of P_t^{bal} and P_t^{scar} to forecast $P_{t+15\text{min}}$.

(iii) Without scarcity-function component. The system imbalance, related price features, and the scarcity knowledge in Eqs. (15)–(16) are not used to construct the latent representation of P_t^{scar}. The model therefore relies only on the latent representations of P_t^{bal} and P_t^{mkt} to forecast $P_{t+15\text{min}}$.

5.1.3. Empirical scaling laws

We further investigate empirical scaling laws with respect to input length and forecasting horizon. At each forecasting time t, the model receives historical features from $t - N$ to t and predicts the target price at $t + M$, where N denotes the look-back window and M denotes the forecasting horizon. We evaluate multiple look-back windows, $N \in \{0, 60, 180, 1440\}$ min, and forecasting horizons, $M \in \{15, 30, 45, 60, 120, 180, 360, 540, 720, 1080, 1440\}$ min. For each (N, M) configuration, we report the testing metrics to characterize how forecasting performance scales with both the amount of historical information and the prediction horizon. In real-time operation, some input signals observed at time t may not be immediately available for forecasting . This can occur due to communication delays, publication lags, or data-fetching latency [31]. Therefore, increasing the forecasting horizon M can also be interpreted as evaluating the effect of **delayed input availability**. For example, when $N = 0$ and $M = 180$, the model uses only the signals available at time t to forecast the imbalance price at $t + 180$ min; equivalently, this reflects a setting in which the most recent available input for forecasting the price at time t is delayed by 180 min.

5.2. Case study results

5.2.1. Model comparison

We compare a set of models to evaluate two aspects of imbalance price forecasting: predictive and computational performance.

Predictive Performance. As shown in Table 3, the three Naïve baselines yield weak forecasting performance, confirming that simple persistence heuristics are insufficient. Compared with the best Naïve baseline, MRINN reduces AQL by **18.63%**, MAE by **12.31%**, and RMSE by **12.06%**. Among lagged-price-based models, XGB and MLP clearly outperform LQR, indicating that the relationship between historical prices and future imbalance prices is nonlinear. However, the performance difference between XGB and MLP is small, suggesting that increasing nonlinear modeling capacity over lagged-price inputs provides limited additional gains. MRINN further reduces AQL by **8.04%** compared with the strongest lagged-price-based baseline. This supports the non-invertibility argument: lagged imbalance prices are compressed market outcomes and may lose information from the raw market signals. Raw-feature-based neural baselines achieve similar AQL, MAE, and RMSE to MRINN, with no statistically significant difference. However, these baselines exhibit quantile crossing, with AQCR values ranging from **6.58%** to **8.53%**. In contrast, MRINN uses a hierarchical head and achieves zero quantile crossing. Fig. 3**(a)–(c)** further shows that the point forecasts of MRINN are mainly concentrated in the dense central region. Extreme prices remain difficult to forecast pointwise, as they are often associated with high uncertainty. Therefore, probabilistic forecasts are more suitable for decision-making in balancing markets, as they provide uncertainty information beyond a single point estimate.

Computational Performance. The main advantage of MRINN lies in its computational efficiency. Although raw-feature-based neural baselines achieve similar predictive performance, they require substantially more parameters (11.3k–28.3k). MRINN uses only **1817** trainable parameters, around **90%** fewer than the competing neural baselines. As shown in Fig. 3**(d)**, MRINN also achieves the lowest training time (0.82 s per epoch) and inference time (1.19 s) among the neural models. Compared with the second-most efficient neural baseline (TimesNet), which requires 9.23 s per epoch for training and 1.43 s for inference, MRINN reduces training time by 91.11% and inference time by 16.61%. This efficiency advantage is particularly important for industrial deployment under operational constraints, where forecasting models may need to run on local servers or edge devices. MRINN achieves this by encoding known rules directly in the model. This makes the model compact without sacrificing predictive accuracy. In contrast, purely data-driven models must use larger model capacity to relearn these rules, resulting in unnecessarily heavier models and higher computational cost.

5.2.2. Price component ablation

Table 4 quantifies the contribution of each price component through a leave-one-component-out ablation study. Removing any component degrades both probabilistic and pointwise forecasting performance, confirming that P_t^{bal}, P_t^{mkt}, and P_t^{scar} provide complementary information. The largest AQL degradation occurs when removing P_t^{scar}, with an **11.84%** increase, indicating that the scarcity-function component contributes most to uncertainty modeling and tail-risk characterization. The largest MAE degradation occurs when removing P_t^{bal}, with a **9.85%**

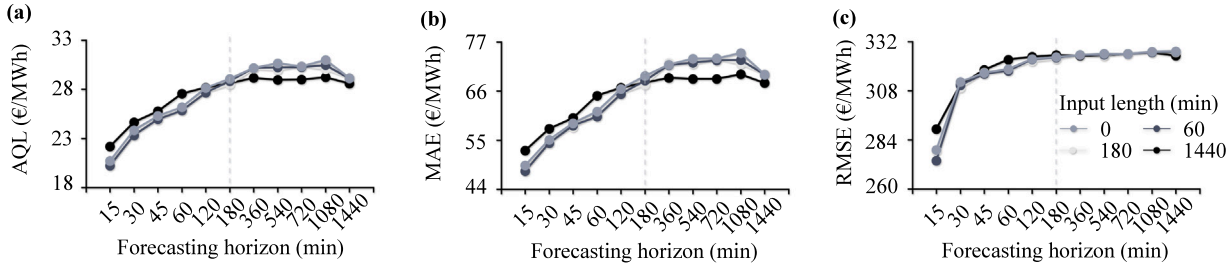


Fig. 4. Empirical results. **(a)–(c)** Scaling-law analysis under different forecasting horizons and input lengths: **(a)** AQL, **(b)** MAE, and **(c)** RMSE.

Table 3
Forecasting performance comparison.

Model	Probabilistic		Pointwise		Params
	AQL	AQCR	MAE	RMSE	
Naïve[1]	25.46	0.00	56.29	321.84	–
Naïve[2]	25.49	0.00	58.51	315.36	–
Naïve[3]	25.44	0.00	58.53	316.77	–
LQR	24.48	0.62	53.30	294.26	14
XGB	22.54	2.01	50.90	294.35	7.7k
MLP	22.51	1.89	50.87	294.02	8.9k
PatchTST	20.72	6.77	49.43	277.73	17.9k
iTransformer	20.71	8.44	49.38	277.32	13.6k
TimesNet	20.70	7.17	49.45	277.47	11.3k
TimeXer	20.73	8.53	49.46	278.08	21.7k
AttnBiLSTM	20.72	6.58	49.34	277.51	28.3k
MRINN	**20.70**	**0.00**	**49.36**	**277.33**	**1.8k**

Table 4
Ablation study of price component.

Ablation	Probabilistic		Pointwise		Rank
	AQL	AQCR	MAE	RMSE	
w/o P_t^{bal}	22.63	0.00	54.22	288.33	2
w/o P_t^{mkt}	22.73	0.00	52.07	311.12	3
w/o P_t^{scar}	23.15	0.00	53.22	310.54	4
All	**20.70**	**0.00**	**49.36**	**277.33**	**1**

increase, suggesting that the balancing-energy component is most important for average pointwise accuracy. The largest RMSE degradation occurs when removing P_t^{mkt}, with a **12.18%** increase, suggesting that the market-reference component is especially important for controlling large pointwise errors and extreme deviations. Overall, the full model achieves the best performance, demonstrating the importance of jointly incorporating all three price components.

5.2.3. Empirical scaling laws

Fig. 4**(a)–(c)** illustrate that metrics degrade as the forecasting horizon increases, confirming that longer-horizon imbalance price forecasting is substantially more challenging. This result is also relevant for industrial real-time deployment, since increasing the forecasting horizon can be interpreted as a proxy for delayed input availability. If the most recent signals are not available, the model must effectively forecast further ahead using older observations. **For short-term forecasting horizons** (below 180 min), shorter input windows (0 and 60 min), tend to achieve the best or near-best performance. In contrast, using a very long input length of 1440 min (1 day) leads to worse performance in this regime, suggesting that distant historical information may introduce noise that is less relevant for short-term prediction. **For longer forecasting horizons** (beyond 180 min), the long input window (1440 min) becomes increasingly competitive. This trend suggests that longer-horizon prediction benefits more from low-frequency temporal structure, such as slower market dynamics and daily patterns. In other words, while recent information dominates short-term prediction, long-range contextual information becomes more valuable as the forecasting target moves further into the future. This finding provides practical guidance for model deployment, as the input design can be adapted to the intended forecasting horizon to balance predictive accuracy and computational efficiency.

6. Conclusion

This paper investigated probabilistic imbalance price forecasting under explicit market rules. The results show that compressing raw market signals into lagged imbalance prices can cause information loss, while raw-feature-based models better preserve the information required for uncertainty-aware forecasting. MRINN further demonstrates that market rules should be used together with neural networks. From an industrial deployment perspective, MRINN supports resource-efficient and computationally sustainable forecasting under operational constraints. The price component ablation confirms the complementary role of different price components. Finally, the scaling-law analysis shows that the optimal input length depends on the forecasting horizon. These results provide practical guidance for designing compact and deployable imbalance price forecasting models in operational energy systems.

A key limitation is that a model trained for one country cannot be directly transferred to another country with different market rules. The general method remains applicable, but the corresponding market rules must be adapted to reflect the target market design. Additionally, if market rules evolve over time, the rules described in Section 3 must be updated accordingly to remain aligned with the prevailing market design. Future work could investigate the reusability of differentiable operators across countries and develop a foundation–model–style framework that integrates general data-driven representations with market-specific rules.

CRediT authorship contribution statement

Runyao Yu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Julia Lin:** Writing – review & editing, Validation, Investigation. **Derek W. Bunn:** Writing – review & editing, Supervision. **Jochen Stiasny:** Writing – review & editing, Supervision. **Wentao Wang:** Writing – review & editing, Investigation. **Yujie Chen:** Writing – review & editing, Investigation. **Tara Esterl:** Resources, Project administration. **Peter Palensky:** Writing – review & editing, Supervision. **Jochen L. Cremer:** Writing – review & editing, Supervision, Resources.

Declaration of Using Generative AI

During the preparation of this work, the author(s) used ChatGPT for language refinement, including checking grammar and improving wording. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

A.1. Hardware description

We evaluate the computational performance on an **Intel Core i7-1265U CPU**. This setting reflects realistic industrial energy trading applications, where forecasting models may be deployed on site-level servers rather than high-performance GPUs. In such environments, model size and inference latency are practical constraints. Therefore, the parameter efficiency of MRINN becomes increasingly relevant when computational resources are further restricted.

A.2. Hyperparameter optimization

For each model, we perform exhaustive hyperparameter optimization using grid search. The best model checkpoint is selected according to the validation performance and saved for final evaluation. The maximum number of training epochs is set to 70 for all neural models. We also evaluate batch sizes of 512, 1024, and 2048. Empirically, smaller batch sizes lead to earlier convergence in terms of epochs but require longer training time, whereas a batch size of 2048 does not consistently reach the best validation loss within 70 epochs. Therefore, we use a batch size of 1024 as a practical trade-off. For invalid hyperparameter combinations, the corresponding configurations are skipped during grid search. For example, in PatchTST, configurations where the patch length exceeds the input length, such as using `patch_len = 4` when the input length is 2, are excluded because they cannot form valid input patches. The full hyperparameter search spaces are summarized in Table 5.

Table 5
Hyperparameter search space.

Model	Search space
MRINN	hidden_size: {8, 32, 128} n_layers: {2, 3, 4}
iTransformer	hidden_size: {8, 32, 128} e_layers: {2, 3, 4} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5}
PatchTST	hidden_size: {8, 32, 128} e_layers: {2, 3, 4} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5} patch_len: {2, 4, 8}
TimesNet	hidden_size: {8, 32, 128} conv_hidden_size: {8, 32, 128} e_layers: {2, 3, 4} dropout: {0.1, 0.3, 0.5}
TimeXer	hidden_size: {8, 32, 128} e_layers: {2, 3, 4} n_heads: {2, 4, 8} d_ff: {8, 32, 128} dropout: {0.1, 0.3, 0.5}
AttnBiLSTM	hidden_size: {8, 32, 128} layers: {2, 3, 4} n_heads: {2, 4, 8}

Data availability

We open source the data and code. Link is provided in the paper.

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